
Special Topics in AI

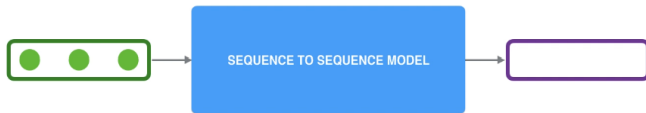
Lecture 2: Attention

University of Mazandaran

Fall 1405

Sequence-to-Sequence (Seq2Seq)

Models that map one sequence of items into another

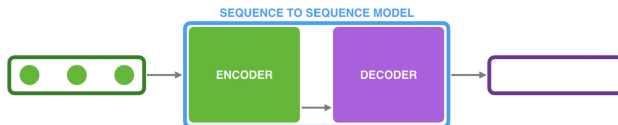


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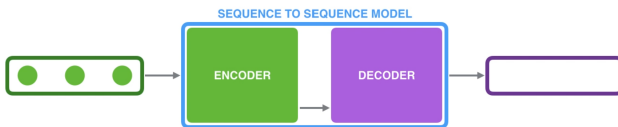


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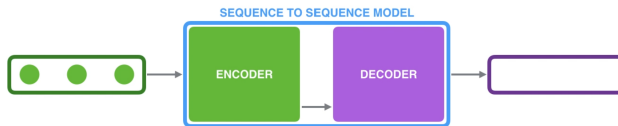
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- ▶ Encoder: takes a variable-length sequence of elements as input and transforms it into a context representation with a fixed-size
- ▶ Decoder: maps the context of a fixed-size to a variable-length sequence of elements

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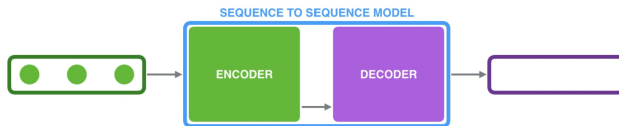
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Success in machine translation, text summarization and image captioning

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Success in machine translation, text summarization and image captioning

Machine translation is the use of computational techniques to translate text or speech from one language to another.

Two important concepts:

1. Alignment: identifying which parts of the input sentence are relevant to each word in the output
2. Translation: using relevant information to select the appropriate output from a set of possible outputs.

Neural Machine Translation With a seq2seq Model

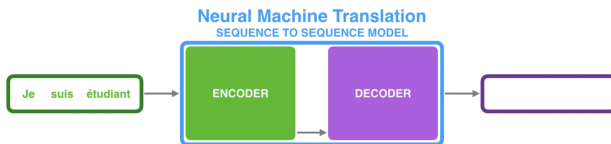
In Neural Machine Translation (NMT), the alignment-translation is done end-to-end using seq2seq model

- ▶ An encoder neural network reads and encodes a source sentence into a fixed-length context vector
- ▶ A decoder then outputs a translation from the encoded context vector

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Defining seq2seq for NMT¹: Encoder

- ▶ An encoder encodes the input sentence, a sequence of vectors $\mathbf{x} = (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N)$, into a context vector

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 - ▶ the encoder is typically an RNN/LSTM

$$\mathbf{h}_t = f(\mathbf{h}_{t-1}, \mathbf{x}_t)$$

For a simple RNN $\mathbf{h}_t = \tanh(\mathbf{W}_x \mathbf{x}_t + \mathbf{W}_h \mathbf{h}_{t-1} + b)$.

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- ▶ Each hidden state summarizes everything seen so far

$$\mathbf{h}_1 = f(\mathbf{h}_0, \mathbf{x}_1) \rightarrow \text{information about } \mathbf{x}_1$$

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- ▶ Finally $\mathbf{c} = q(\{\mathbf{h}_1, \dots, \mathbf{h}_N\})$ where
 - ▶ $\mathbf{c}$ is the context vector generated from the hidden states
 - ▶ q is any linear or nonlinear function

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$$p(\mathbf{y}) = \prod_{t=1}^T p(\mathbf{y}_t \mid \{\mathbf{y}_1, \dots, \mathbf{y}_{t-1}\}, \mathbf{c}), \text{ where } \mathbf{y} = (\mathbf{y}_1, \dots, \mathbf{y}_T)$$

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- ▶ With an RNN decoder, each conditional probability is modeled as:

$$p(\mathbf{y}_t \mid \{\mathbf{y}_1, \dots, \mathbf{y}_{t-1}\}, \mathbf{c}) = g(\mathbf{y}_{t-1}, \mathbf{s}_t, \mathbf{c})$$

- ▶ g is a nonlinear (multi-layered) function
- ▶ $\mathbf{s}_t$ is the hidden state of the RNN

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The Issue With The seq2seq Models

- ▶ The model needs to compress all necessary information of a source sequence into the fixed-length context vector $\mathbf{c}$
- ▶ It is challenging for the model to compress all information into a fixed-length context, especially when dealing with long sequences.

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- ▶ The attention mechanism¹ overcomes the bottleneck issue of encoder-decoder
- ▶ Intuitively it allows the model to **focus** on relevant parts of the input, while decoding the output
- ▶ In the context of NMT: Each time the model generates a word, it searches for a set of positions in the source sentence where the most relevant information is concentrated.

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An intuitive example

In the following sentence, determine the word to which "it" refers.

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This mental weighting of relevant words is what attention does!

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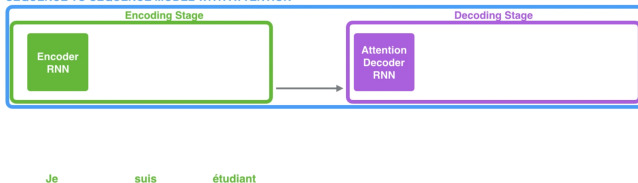
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Attention

- ▶ The encoder-decoder with attention does not need to encode the whole input into a single fixed-length vector
- ▶ It instead encodes the input into a sequence of vectors
- ▶ Then it choose a subset of these vectors adaptively while decoding the output
- ▶ This frees the model from having to **squash all the information** of the source into a fixed-length vector.

Neural Machine Translation

SEQUENCE TO SEQUENCE MODEL WITH ATTENTION



Step-by-step

1. **Input encoding**: each token (say word!) i from the input sentence is encoded by encoder to a hidden vector $\mathbf{h}_i$

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 - ▶ when scoring the first output for the decoder, $\mathbf{s}$ will be a vector of zeros!

Alignment score and alignment function

- ▶ The alignment score e_{ti} indicates how relevant each source position i is when generating the current target token $\mathbf{y}_t$
- ▶ It is computed using a small feed-forward neural network represented by $a(\cdot)$:

$$e_{ti} = a(\mathbf{s}_{t-1}, \mathbf{h}_i) = \tanh(\mathbf{W}_a \mathbf{s}_{t-1} + \mathbf{U}_a \mathbf{h}_i)$$

- ▶ The alignment model is trained jointly with all the other components of the model

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$$\alpha_{ti} = \text{softmax}(e_{ti})$$

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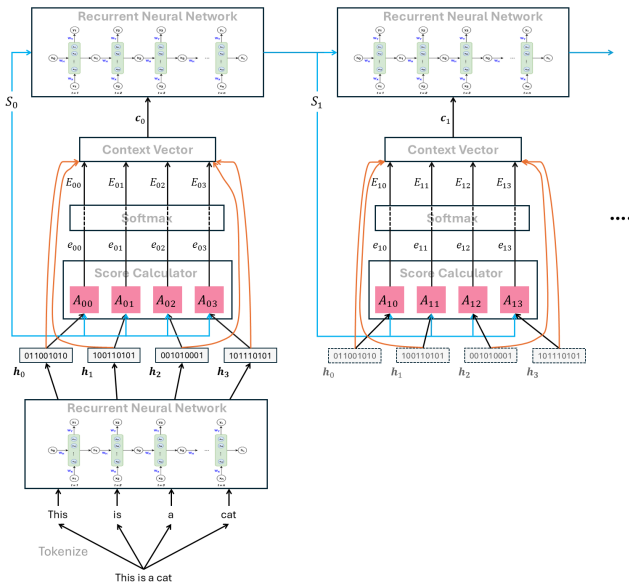
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4. **Context vector**: a unique context vector $\mathbf{c}_t$ is generated by a weighted sum of all encoder hidden states weighted by the weights computed in previous step

$$\mathbf{c}_t = \sum_{i=1} \alpha_{ti} \mathbf{h}_i$$

Attention



- ▶ **Issue:** computing the attention score is cumbersome!
 - ▶ say, the encoder sequence is of length m and the decoder sequence is of size n
 - ▶ we require $m \times n$ alignment score calculation (evaluating a small feed-forward network)
 - ▶ the decoder is also an RNN, so to compute s_i , we have to compute s_{i-1} first!
 - ▶ the $m \times n$ alignment score calculation could not be paralleled

Do we really need the recurrent state $\mathbf{s}_i$ at all?

²Attention is All You Need, Vaswani et al., NeurIPS (2017)

Do we really need the recurrent state s_i at all? No!

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Attention: A More Efficient Version²

1. Pass each input token $\mathbf{x}_i$ through the encoder to get $\mathbf{H}^{\text{enc}} = [\mathbf{h}_1^{\text{enc}}, \mathbf{h}_2^{\text{enc}}, \dots, \mathbf{h}_N^{\text{enc}}]$
2. Pass each output token $\mathbf{y}_j$ through the decoder to get $\mathbf{H}^{\text{dec}} = [\mathbf{h}_1^{\text{dec}}, \mathbf{h}_2^{\text{dec}}, \dots, \mathbf{h}_T^{\text{dec}}]$
3. Project these representation into three learned spaces:
 $\mathbf{Q} = \mathbf{H}^{\text{dec}} \mathbf{W}_Q$, $\mathbf{K} = \mathbf{H}^{\text{enc}} \mathbf{W}_K$, $\mathbf{V} = \mathbf{H}^{\text{enc}} \mathbf{W}_V$
4. Compute all pairwise score with one matrix multiplication!
 $\mathbf{E} = \mathbf{QK}^T$
5. Scale the attention scores: $\mathbf{E} = \frac{\mathbf{QK}^T}{\sqrt{d_k}}$
 - scaling prevents large dot products from pushing the softmax into regions with very small gradients
6. Convert scores to attention weights: $\mathbf{A} = \text{softmax}\left(\frac{\mathbf{QK}^T}{\sqrt{d_k}}\right)$
7. Generate the output representation: $\mathbf{Z} = \mathbf{AV}$.

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The General Attention Mechanism

Makes use of 3 main components: **Queries**, **Keys**, and **Values**.

This is similar to information retrieval systems:

- ▶ Search for a video on Youtube!
- ▶ **Query**: text in the search bar
- ▶ **Keys**: video titles, descriptions, etc., associated with candidate videos
- ▶ **Values**: best matched videos